

MAHADI HASAN TANMAY

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EDUCATION

B.Sc. in Computer Science and Engineering, United International University, Dhaka

Fall 2022 - Expected Summer 2027 | CGPA: 3.53 / 4.00

Merit scholarships (Spring 2026, Spring 2023); academic waivers (Summer 2023, Spring 2023, Fall 2022)

HSC 2020, Bhola Government College (Science), GPA 5.00 | SSC 2018, Bhola Abdur Rob High School and College (Science), GPA 4.61

RESEARCH

Undergraduate Thesis (in progress): Agentic AI Security in Industrial Automation and Cyber-Physical Systems

Supervisor: Azizur Rahman Anik, Lecturer, Department of CSE, UIU

- Investigating how autonomous agentic AI deployed for predictive maintenance and smart-grid control introduces physically consequential attack surfaces, including data poisoning, prompt injection, and adversarial manipulation.
- Empirically evaluating production agentic frameworks against a defined set of industrial attack scenarios to identify where current defences fail.
- Addresses a gap between LLM/agent security research, which rarely covers industrial control systems, and ICS anomaly detection, which rarely assumes an adversarial or agentic threat model.

Final Year Design Project (FYDP I-III), UIU - Three-trimester capstone sequence forming the applied track of the thesis above; FYDP-I in progress.

PROJECTS

Smart Hospital Queue Management System | Solo | Summer 2026 | React 19, FastAPI, MongoDB

- Built a full-stack outpatient token and queue system with four roles (guest, patient, doctor, admin), around 35 REST endpoints across 9 routers, and 9 MongoDB collections.
- Enforced queue limits, no-show timeouts, and role permissions server-side rather than in the UI; JWT authentication with bcrypt hashing and revalidation of account status on every request.
- github.com/mahadi2002/smart-hospital-queue-management-system

NeuroVerse - AI-Integrated Student Wellness Platform | Team of 5 | Spring 2026

- Owned the authentication and anonymous peer-forum modules; served as backend coordinator and tester for a Node.js/Express and FastAPI hybrid architecture on MongoDB.
- Integrated a FastAPI ML service performing sentiment and crisis analysis via Llama-3.3-70B (Groq), with SentenceTransformer embeddings and NetworkX similarity graphs for content recommendation.
- Deployed at neuro-link-rouge.vercel.app

Railway Passenger Information System (HCI) | Team lead, 6 members | Summer 2026

- Led a user-centred design study of Bangladesh Railway passenger information; designed the semi-structured interview protocol and ran interviews and non-participant observation with 20-25 participants at Dhaka Airport Railway Station.
- Produced the high-fidelity Figma prototype and authored the final report, covering live train tracking, announcement transcripts, seat visualisation, and a Bangla/English toggle.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C

Web: React, Vite, Node.js, Express, FastAPI, REST API design, JWT authentication, Tailwind CSS, Bootstrap

AI / ML: scikit-learn, Hugging Face Transformers, SentenceTransformers, Groq API (Llama 3.3), NetworkX; PyTorch and TensorFlow (familiar)

Security & Networking: Wireshark, Nmap, network protocol analysis

Data & Tools: MongoDB, MySQL, PostgreSQL, Git, Docker, Linux, Vercel, Render, Figma

LEADERSHIP AND ACTIVITIES

UIU Robotics Club - Photography Team Member | 2023 - Present

- Document club events and competitions, producing the visual record used in club communications.

CERTIFICATIONS AND ADDITIONAL INFORMATION

Certifications: Multiple professional courses completed through MyGP Academy

Languages: Bangla (native), English (professional working proficiency)

References: Available on request